



OPEN Mixture-of-experts graph transformers for interpretable particle collision detection

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The Large Hadron Collider at CERN produces immense volumes of complex data from high-energy particle collisions, demanding sophisticated analytical techniques for effective interpretation. Neural Networks, including Graph Neural Networks, have shown promise in tasks such as event classification and object identification by representing collisions as graphs. However, while Graph Neural Networks excel in predictive accuracy, their “black box” nature often limits their explainability, making it difficult to trust their decision-making processes. In this paper, we propose a novel approach that combines a Graph Transformer model with Mixture-of-Expert layers to achieve high predictive performance while embedding explainability into the architecture. By leveraging attention maps and expert specialization, the model offers insights into its internal decision-making, linking predictions to physics-informed features. We evaluate the model on simulated events from the ATLAS experiment, focusing on distinguishing rare Supersymmetric signal events from Standard Model background. Our results highlight that the model achieves competitive classification accuracy while providing interpretable outputs that align with known physics, demonstrating its potential as a robust and transparent tool for high-energy physics data analysis. This approach underscores the importance of explainability in machine learning methods applied to high energy physics, offering a path toward greater trust in AI-driven discoveries.

The Large Hadron Collider (LHC) at CERN offers an exceptional setting for investigating the fundamental constituents of matter. Experiments like ATLAS¹, generate extensive data streams from high-energy particle collisions, enabling the study of particle interactions across a wide energy spectrum. While this data is invaluable for advancing particle physics, the sheer scale and complexity of the information present significant analytical challenges². For instance, ATLAS event reconstruction processes data at rates exceeding 3.5 terabytes per second³, requiring advanced computational techniques to extract meaningful insights.

In this work, we leverage simulated LHC data as a controlled benchmark to address a key challenge in modern high-energy physics (HEP): the development of interpretable machine-learning models. Simulated data, where the underlying physics is fully known, provides an ideal environment to evaluate the trustworthiness of machine-learning methods by verifying whether their outputs align with established physical principles. Machine learning (ML) and deep learning techniques have demonstrated strong potential in tackling HEP challenges⁴, including particle identification⁵, event reconstruction⁶, and background subtraction⁷. Nevertheless, these methods often operate as “black boxes”, making it difficult to interpret their decision-making processes. This lack of transparency is a critical limitation, particularly in fields like HEP, where explainability is essential for establishing confidence in novel discoveries.

From a performance point of view, Graph Neural Networks (GNNs) have emerged as promising tools for analyzing complex datasets like those produced by the LHC. By representing particles and their interactions as nodes and edges in a graph⁸, GNNs can model intricate relationships that other methods might overlook. While GNNs have demonstrated state-of-the-art performance in classification tasks, their value extends beyond predictive accuracy. A major focus of this work is on developing GNN-based models that inherently provide

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insights into their internal decision-making processes, ensuring that their predictions align with physical intuition. The terms *explainability* and *interpretability* are often used interchangeably in the literature, although some works attempt to distinguish between them. However, there is currently no universally accepted consensus within the Explainable Artificial Intelligence (XAI) community on how to clearly differentiate these concepts^{9–12}. Given this ongoing debate and lack of standardization, in this work we adopt the term explainability to refer broadly to the overall process of making AI systems more understandable to humans.

Despite advances in these areas, GNNs often remain “black box” models. To address this issue, various post-hoc explainability methods have been introduced for graph neural networks¹³, such as PGExplainer¹⁴ and GNNExplainer¹⁵, alongside data attribution techniques¹⁶. These approaches aim to provide insight into how GNNs make their predictions by identifying important input features or graph substructures. Nonetheless, while these methods offer some understanding, they still fall short of providing a comprehensive understanding of the model’s internal workings^{17,18}.

Our work addresses these limitations by proposing a novel model that integrates explainability directly into its architecture. Specifically, we build upon the Graph Transformer (GT) model¹⁹. This architecture offers the possibility of visualizing attention maps, enabling us to illustrate how the model identifies relationships within graph structures. By examining these attention maps, we can verify whether the model focuses on graph regions consistent with known physics. To further enhance explainability, we replace the standard feed-forward layer of the Transformer with a Mixture of Experts (MoE) layer²⁰. This design enables the model to distribute its outputs across specialized “experts,” each targeting distinct data subsets, thereby providing a clearer understanding of how individual components contribute to the final decision. This enhanced transparency allows us to trace the model’s reasoning pathways and verify their alignment with physical principles. We test our approach on Monte Carlo-simulated collision events, targeting hypothetical particles predicted by Supersymmetry (SUSY). These datasets include both rare SUSY signal events and dominant Standard Model (SM) backgrounds, following established ATLAS methodologies²¹. The classification task, distinguishing rare signal from high-rate background, mirrors the challenges typical in HEP. Our focus is not solely on achieving high performance but on validating the explainability of the model. By leveraging this controlled dataset, we demonstrate how our method identifies critical features and regions of the graph that align with the expected physics, fostering trust in the model’s predictions. In this work, we emphasize that the explainability of machine-learning models is as crucial as their accuracy, particularly when exploring the potential for new physics. Our approach illustrates how explainable GNN-based methods can provide high predictive accuracy, reinforcing the connection between computational predictions and physical intuition. This focus on explainability ensures that our methodology is not only a powerful analytical tool but also a reliable framework for future discoveries in particle physics.

Related works

Graph neural networks and graph transformers

GNNs are a specialized class of neural networks designed to process and analyze graph-structured data, with their key strength lying in their ability to learn representations that capture complex relationships among nodes in a graph. GNNs are extremely versatile and can be adapted to various graph-related tasks, including node and graph classification, regression, edge prediction, graph generation, and node clustering. They have found wide-ranging applications in particle physics^{8,22} and HEP, especially in particle tracking and reconstruction^{23,24}. The physics tasks at the LHC experiments have provided many potential applications where graph neural networks have been successfully applied²⁵. Building upon the foundation of GNNs, GTs have recently emerged as a powerful graph learning method. GTs draw inspiration from the success of Transformers in natural language processing (NLP)²⁶ and computer vision (CV)²⁷. GTs excel at handling dynamic and heterogeneous graphs, utilizing both node and edge features¹⁹. Various adaptations and expansions of GTs have demonstrated their superiority in addressing multiple challenges in graph learning, such as processing large-scale graphs²⁸. In addition to these advancements, explainability methods for attention mechanisms have provided valuable insights into the inner workings of self-attention models, further enhancing their explainability²⁹.

Mixture of experts

The Mixture of Experts (MoE²⁰) architecture offers a modular approach to scaling neural networks size. As models increase in size and complexity, they require substantial computational resources. MoE addresses this challenge by introducing expert specialization, where multiple subnetworks, or “experts,” are trained to handle different regions of the input space. Instead of using a single, large model for all inputs, MoE incorporates a gating mechanism that dynamically assigns each input to a subset of experts, selecting the most relevant ones based on learned patterns²⁰. This mechanism allows for task-specific specialization, allowing each expert to focus on different aspects of input data³⁰. As a result, the architecture can scale more efficiently, allowing for higher model capacity without the need for proportionally increased computation for every input. By activating only a small number of experts at a time, MoE provides a computationally efficient solution for handling large and complex models³¹. Moreover, recent advancements in explainability for MoE architectures, such as analyzing expert utilization and routing behavior, have deepened our understanding of their decision-making processes and modality-specific specializations³².

Explainability for high energy physics

Several explainability methods have been applied to HEP tasks, illustrating their utility across different domains. Models like XGBoost³³ and Deep Neural Networks (DNNs) have been interpreted using SHapley Additive exPlanations (SHAP)³⁴. SHAP studies have proven effective to understand how signal events, characterised by the presence of a Higgs boson candidate, are distinguished from background noise through physically relevant kinematic variables³⁵. Layerwise Relevance Propagation (LRP)³⁶ has been used to enhance the explainability of

DNNs for classifying boosted Z boson jets³⁷. Through LRP-based heatmaps, the method highlighted relevant detector regions. The application of LRP has also been extended to GNNs in the particle-flow reconstruction task³⁸. By distributing relevance scores, the study quantified the importance of input features such as charge and energy, as well as the connections within graph structures. Additionally, data attribution techniques like TracIn³⁹ have been employed to identify impactful training data in the ATLAS datasets considered in our paper²¹. This approach improved both model efficiency and explainability by refining training datasets and revealing influential patterns¹⁶. However, these methods often exhibit brittleness and lack reliability in challenging real-world scenarios. For instance, studies have shown that many post hoc explainability techniques fail fundamental sanity checks, raising concerns about their validity and robustness¹⁸. Other research has highlighted the fragility of interpretations, particularly when faced with slight changes in input or model parameters¹⁷. Attribution-based methods, in particular, have been criticized for being overly sensitive to model biases, making their insights less reliable in practical applications⁴⁰. To address these challenges, we propose intrinsic techniques that integrate explainability directly into the model architecture and learning process, enabling the analysis of the model's internal decision-making mechanisms for complex HEP tasks without relying on post-hoc explanations.

Experimental setup

Dataset

The SUSY dataset, derived from the ATLAS publication²¹ and available in the CERN open data repository (<https://opendata.cern.ch/record/28100>), contains Monte Carlo simulated collision events for a set of signal models (Signal) and SM backgrounds – where the two major sources are hereby considered (Background). Details on the two types of events are given below:

- **Signal:** Collision signal events are depicted in Fig. 1. Two parent particles (named chargino, $\tilde{\chi}_1^\pm$, and next-to-lightest neutralino, $\tilde{\chi}_2^0$) are produced at the collision point and promptly decay into a W and Higgs boson, respectively, and the dark matter candidate (the lightest-neutralino, $\tilde{\chi}_1^0$). The W boson decay immediately into a lepton (l) and a neutrino (ν), and the Higgs boson in a pair of b-quarks.
- **Background:** SM background events represent the processes of pair production of two top quarks ($t\bar{t}$) and the production of a top quark and a W boson (Wt or single top). Other sources of SM backgrounds considered in the ATLAS paper are not considered here as potentially more distinguishable from signal and anyway contributing for less than 20% of the total background. Both top-quark processes mimic the signal as they result in similar final particles.

Signal models are parametrised as a function of two variables, the mass of the SUSY parent particles (equal, according to the model) and the mass of the dark matter particle. The mass value ranges between 125 GeV (the value of the Higgs mass) and 1000 GeV for the former, and 0 and 400 GeV for the latter, although the highest statistics models are in the region where the mass splitting between parent and dark matter particles is close to the Higgs mass, as directly targeted by the ATLAS paper. The kinematic of the signal events depends on the choices of masses, and each choice represents a different model. If SUSY were realised in nature, only one choice of mass parameters would be the correct one. Since this is unknown a priori, all models are mixed together with equivalent probability.

The main task involves detecting the rare signals amidst a large background of SM processes. In real data, signal events are expected to be well below 0.1% of the total background, even after dedicated selections are

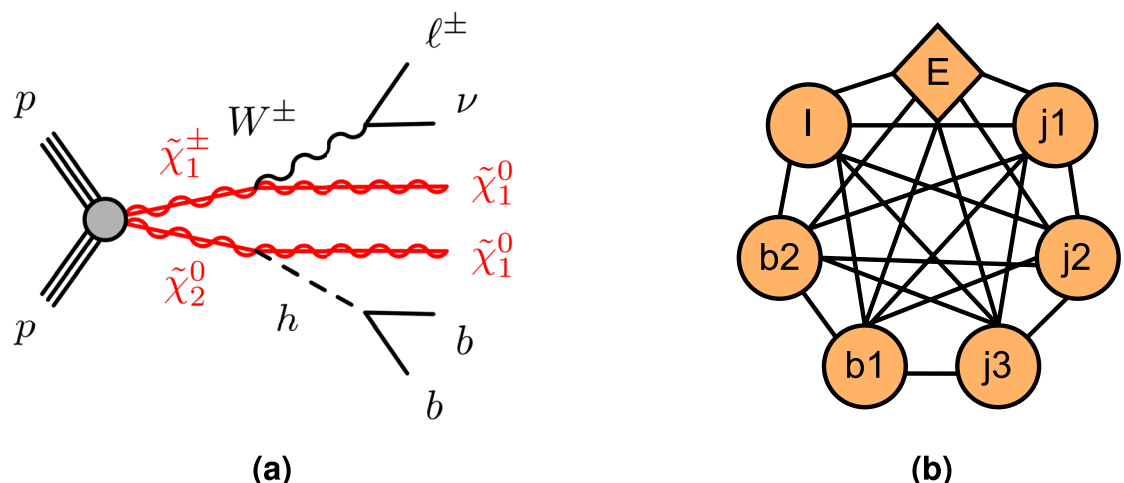


Fig. 1. (a) Diagram of the SUSY signal process, showing chargino and neutralino decaying into W and Higgs bosons, with leptons, neutrinos, and b-quarks in the final state. (b) Schematic representation of a particle collision event modeled as a fully connected graph. Nodes represent reconstructed particles and objects: jets (j1, j2, j3), b-tagged jets (b1, b2), lepton (l), and missing transverse energy (E). The diamond shape for E indicates that it is not a directly reconstructed particle but an inferred quantity.

Particle	F1	F2	F3	F4	F5	F6
$j1$	p_T^{j1}	η_{j1}	ϕ_{j1}	q_{j1}	-	-
$j2$	p_T^{j2}	η_{j2}	ϕ_{j2}	q_{j2}	-	-
$j3$	p_T^{j3}	η_{j3}	ϕ_{j3}	q_{j3}	-	-
$b1$	p_T^{b1}	η_{b1}	ϕ_{b1}	q_{b1}	m_{b1}	-
$b2$	p_T^{b2}	η_{b2}	ϕ_{b2}	q_{b2}	m_{b2}	-
$lepton$	p_T^l	η_l	ϕ_l	-	-	-
$energy$	E_T^{Miss}	-	ϕ_{ETMiss}	-	-	σ_{ETMiss}

Table 1. Kinematic properties of particles.

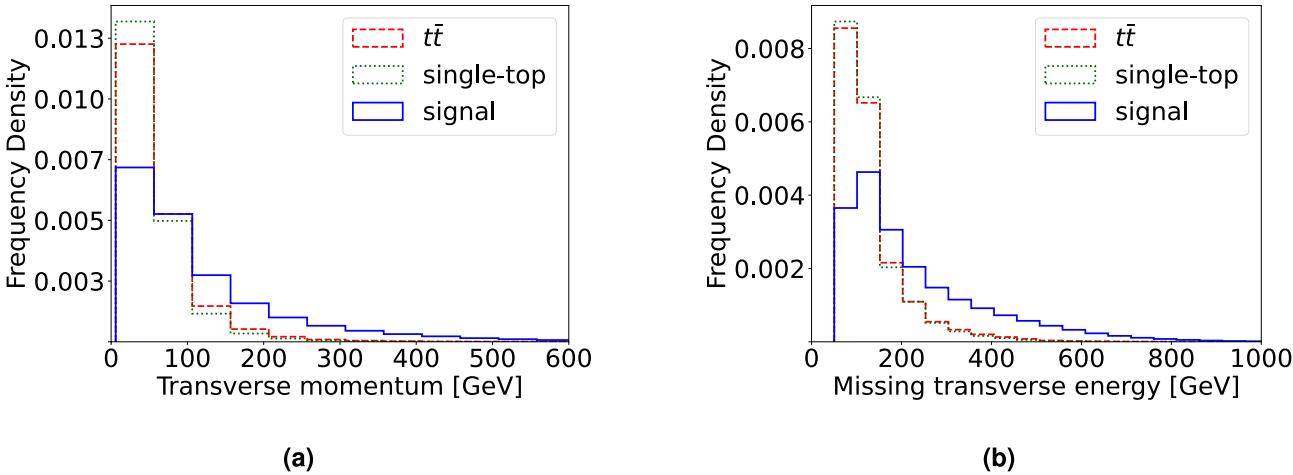


Fig. 2. Distributions of (a) p_T of lepton and (b) for the E_T^{Miss} , comparing signal and the two main background processes.

imposed. To distinguish these events, we rely on kinematic features that offer significant discriminative power. We represent each collision event as a fully connected graph G with 6-7 nodes (N), where each node represents a particle and contains up to 6 kinematic features. Mathematically, each event is encoded as a feature matrix $X \in \mathbb{R}^{N \times 6}$, where rows correspond to particles and columns to their features. The scheme for particle collisions and their corresponding features are depicted in the graph in Fig. 1b and, in detail, in table 1. With reference to the Table, each row represents a distinct particle that was reconstructed in the detector. The first rows correspond to the jets ($j1$, $j2$, $j3$), where jets are produced by the hadronisation process of quarks. The $b1$ and $b2$ represent the jets originated from the b-quarks present in signal and background events, and are a subset of the generic jets. For events with 6 nodes, $j1$ and $j2$ are equivalent to $b1$ and $b2$. Events with 7 nodes are those with an extra jet, not depicted in Fig. 1. This categorization is made as the presence of a third jet is expected to offer additional discriminative power. The $lepton$ row, l , refers to either a muon or an electron, depending on the particle identified. Finally, the $energy$ row (E) represents the so-called “missing” energy, used as a proxy for particles invisible to the detector such as neutrinos and dark matter candidates. The dataset columns describe various physical properties of the particles. Feature 1, p_T , represents the transverse momentum magnitude for particles, while for the $energy$ row, it is the E_T^{Miss} quantity (called missing transverse energy). Features 2 and 3 are the pseudorapidity (η , related to the angle of a particle relative to the beam axis θ in collision events) and the azimuthal angle (ϕ) of jet and lepton objects, respectively. Only the azimuthal angle of E_T^{Miss} can be defined due to the unknown longitudinal momentum in head-on collisions at the LHC. Feature 4 is referred to as “quantile” and quantifies the likelihood for a jet to be identified as originating from a b-quark or not according to the ATLAS b -tagging identification procedure. Feature 5, $b1m$ and $b2m$, denote the mass of the b-jets calculated from jet component four-vectors. The final column is σ_{ETMiss} , which is implemented as a feature of the missing transverse energy quantity and directly related to its resolution²¹. In Fig. 2 (a) and (b), key features are shown for signal and the two kind of background events considered here, normalised to the unit area. Top-pair and single-top backgrounds are added together as kinematically similar. All signal models are mixed together, with models defined by (low) high mass particles contributing mostly to the bulk (tails) of the distributions.

Mixture of experts graph transformer

Given a graph G , representing a collision event as explained in Sect. “Dataset”, our task is to predict a label y such that $y = f(G)$. Specifically, the label y encodes the class of the collision event that we aim to predict – *Signal* or *Background* – using a neural network model f . To ensure that the predictions are as interpretable as possible, we design the model f with intrinsic explainability in mind and model it as a GT architecture¹⁹ enhanced with a MoE. The GT processes graph-structured data using self-attention and feed-forward networks, while the MoE enables parameter subsets to specialize in distinct data aspects. This architecture ensures intrinsic explainability by using attention to quantify node relationships. Fig. 4 illustrates the model, highlighting attention maps and the MoE’s role in capturing dependencies and specialization.

We test our proposed model on the Monte Carlo simulated collisions events for signal and SM background events described above. In this setup, each graph event G is characterized by a feature matrix $X \in \mathbb{R}^{N \times 6}$, where N is the number of nodes in the graph and each row vector x_i represents a node embedding with its features. The GT updates nodes representations using a sequence of layers consisting in a self-attention mechanism followed by a feed-forward block. After the initial addition of Laplacian positional encodings^{41–43}, for each node x_i , the attention mechanism computes a weighted sum of value vectors V , where the weights are determined by scaled dot-product attention¹⁹. Let Q, K, V denote the query, key, and value matrices derived from the input X via learned linear projections W_Q, W_K, W_V . The attention for x_i is:

$$\text{Attention}(x_i) = \text{softmax} \left(\frac{q_i K^\top}{\sqrt{d_k}} \right) V \quad (1)$$

where $q_i = x_i W_Q$ is the query vector for x_i , $K = X W_K$ is the matrix of key vectors, $V = X W_V$ is the matrix of value vectors, and d_k is the dimensionality of the key vectors used for scaling. We refer to $\text{softmax} \left(\frac{q_i K^\top}{\sqrt{d_k}} \right)$ as “attention weights”, as they serve to encode the relationships between nodes, effectively quantifying their interactions and dependencies. This view offers an interpretable framework for understanding how the model processes information and makes decisions, which we describe more in detail in Sect. “Analysis of the attention heads”. The GT employs multi-head attention, extending the attention mechanism by employing H independent attention heads, each with its own set of projections W_Q^h, W_K^h, W_V^h for $h = 1, \dots, H$. The outputs of these heads are concatenated and linearly projected back to the original dimensionality. For x_i , the multi-head attention is:

$$\text{MultiHead}(x_i) = \text{Concat}(\text{head}_1, \dots, \text{head}_H) W_O, \quad \text{with} \quad \text{head}_h = \text{softmax} \left(\frac{q_i^h (K^h)^\top}{\sqrt{d_k}} \right) V^h \quad (2)$$

where $q_i^h = x_i W_Q^h$, $K^h = X W_K^h$, and $V^h = X W_V^h$. The output projection W_O combines the H heads into a single representation. After updating the node representations according to Eq. 2, the node embeddings are passed through a feed-forward network (FFN), consisting in a sequence (usually two) learnable linear projections with an intermediate nonlinearity:

$$x_i \mapsto W_2(\text{ReLU}(W_1(x_i) + \beta_1)) + \beta_2 \quad (3)$$

where x_i represents a generic node embedding, $W_1, \beta_1, W_2, \beta_2$ represent learnable weights and biases, and ReLU is a nonlinear activation function.

In order to enhance explainability, we also replace the original FFN with a MoE block, as described next. The MoE architecture consists of n expert networks, denoted as $E_1, \dots, E_n$, and a routing network r . Each expert is a neural network defined as in Eq. (3) with its own parameters. The routing network r determines a set of weights for the expert outputs based on the updated node representation, thus assigning each node of the graph to one or more expert networks. The output y of the MoE block is computed as a weighted combination of the expert outputs:

$$y = \sum_{i=1}^n r(x)_i E_i(x).$$

The MoE enhances explainability by allowing us to identify which experts contribute most to the prediction for a given node, providing insights into the model’s decision-making process. Additionally, the MoE optimizes computational efficiency by leveraging the sparsity of the gating network’s output. When $r(x)_i = 0$, we bypass the computation of $E_i(x)$, thus saving computational resources. We follow (Shazeer et al. 2017)²⁰ for the design of the routing network, and implement noisy *top-K* gating with a load balancing loss to prevent imbalanced routing choices that would lead to the training of a small subset of experts, a well-known issue for MoE models. Details about the routing are given in S1 of Supplementary Materials. The overall architecture is further depicted in Fig. 3 and includes the GT stack and the classification head tailored to the task.

Results

We conducted experiments to evaluate the performance of four architectures—Graph Convolutional Network (GCN), MLP, GT, and our model described in Sect. “Mixture of experts graph transformer”, the Mixture of Experts Graph Transformer (MGT)—on the SUSY dataset. The GCN architecture, uses two convolutional layers with hidden channels of size 80 and 40, respectively. It employs global mean pooling for aggregation and

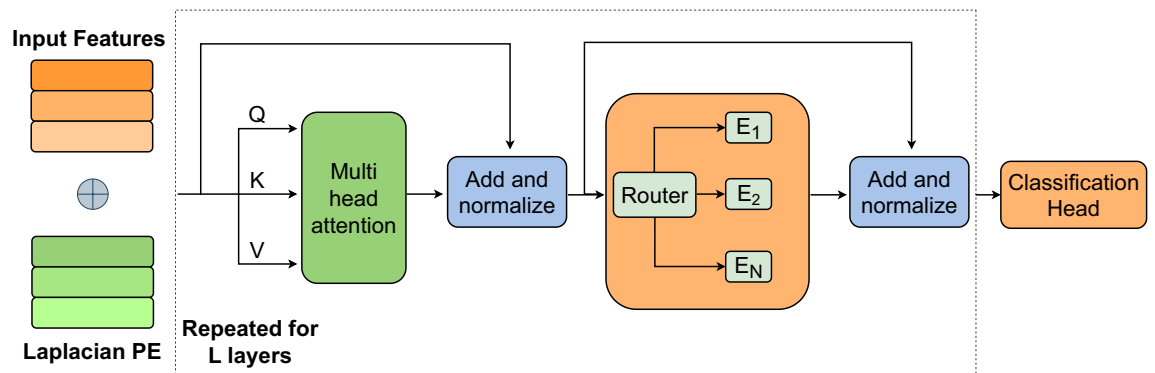


Fig. 3. Detailed illustration of the proposed architecture incorporating multi-head attention with an MoE block. The model begins with Laplacian positional encoding for the input features, followed by multi-head attention and normalization. This structure is repeated across L layers. The MoE block, driven by a gating network, assigns inputs dynamically to specialized expert networks. A classification head processes the final representation to produce predictions.

Model	Accuracy	Precision	Recall	F1	AUC
GCN	0.750 ± 0.0022	0.779 ± 0.2549	0.700 ± 0.0427	0.736 ± 0.0131	0.832 ± 0.0134
MLP	0.829 ± 0.0015	0.826 ± 0.0054	0.835 ± 0.0101	0.830 ± 0.0027	0.913 ± 0.0017
GT	0.849 ± 0.0059	0.850 ± 0.0062	0.849 ± 0.0059	0.849 ± 0.0060	0.928 ± 0.0057
MGT	0.852 ± 0.0005	0.851 ± 0.0008	0.849 ± 0.0016	0.854 ± 0.0004	0.929 ± 0.0039

Table 2. Performance comparison of different architectures (MLP, GCN, GT, and MGT) on the SUSY dataset. Results are reported as the mean $\pm$ standard deviation over ten runs with different random seeds. Metrics include Accuracy, Precision, Recall, F1 score, and AUC.

concludes with a linear layer to output the class predictions. The GCN results show promising differentiation between signal and background, aligning with key kinematic features of the dataset, with detailed analysis, including feature importance and event correlations, provided in S4 of Supplementary Materials. The MLP model also consists of two hidden layers, with dimensions 80 and 40, respectively, and takes the input graph as a flattened vector. The GT architecture is composed of two layers, each with 2 attention heads, Laplacian positional embeddings of size 4 and a hidden size of 80. The MGT model builds upon the GT architecture with a hidden size of 80, 2 attention heads, 6 expert networks each with an expert size of 20, and 2 layers. It also employs a gating mechanism with a weight load factor of 1 for expert selection.

The experiments were conducted using ten runs, with a different random seed for each run, to ensure robust evaluation of model performance. The results, quantified in terms of accuracy, precision, recall, F1-score, and AUC are reported in table 2. The MoE architecture outperforms the other models, achieving superior accuracy and overall performance metrics. This highlights the effectiveness of MoE in leveraging expert networks for enhanced predictive capabilities. Further analysis on training is provided in S3 of the Supplementary Materials.

Discussion

The attention mechanisms and expert specialization in the MoE model provide valuable insights into its explainability and performance. This section discusses the detailed behavior of attention scores and the unique specialization patterns exhibited by expert networks. The overall architecture of the proposed Transformer-based model is illustrated in Figure 4, providing a detailed view of how the graph input is processed through the MoE blocks.

Analysis of the attention heads

Attention scores provide significant insights into the model's internal mechanisms by refining node and feature representations while capturing relationships and dependencies within the graph. In this process, attention acts as a guiding mechanism, directing how information is aggregated and propagated through the network. In an attention map, the attention weights are organized such that each row corresponds to a query (i.e., the element doing the attending), and each column corresponds to a key (i.e., the element being attended to). To interpret the model, we conduct a detailed analysis of the attention scores by examining multiple subsets of the dataset, including the training set, test set, as well as correctly classified signal events (true positives) and correctly classified background events (true negatives). For each of these groups, we compute the attention matrix for each head, averaging the values across the data to uncover global trends in attention behavior. The images shown in the following (Fig. 5, Fig. 6, Fig. 7) are divided into two columns, with 6-node graphs on the left and 7-node graphs on the right. Each row represents the attention given by a specific node to all other nodes in the graph.

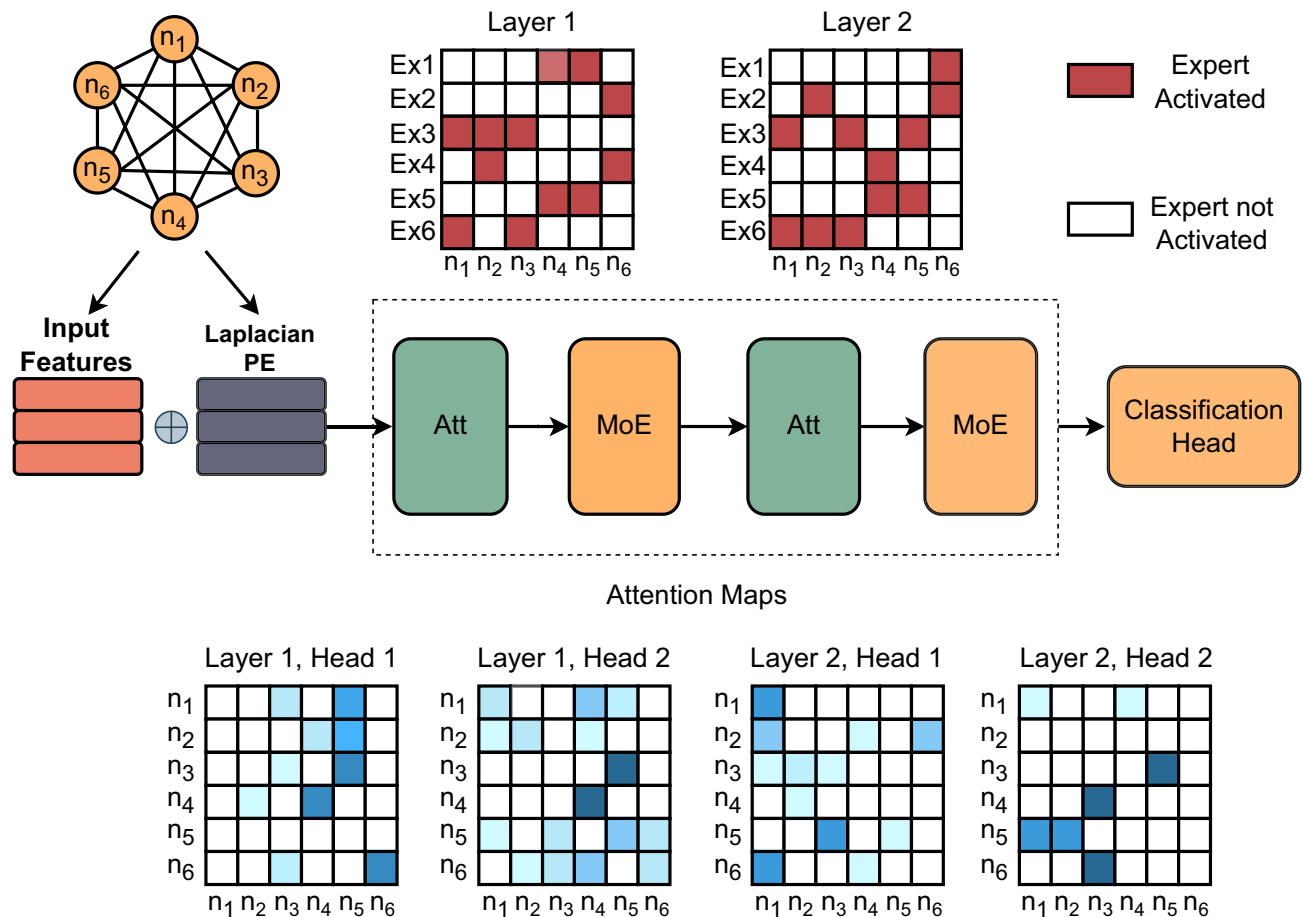


Fig. 4. Overview of the proposed Transformer-based model architecture: the image illustrates an example of the model, which takes as input a graph. The graph is processed by the model, comprising various blocks: **Att**, which is the Multi-Head Attention block, and **MoE** blocks. The visualization includes attention maps derived from the Multi-Head Attention mechanism and the activation patterns of the experts for a single collision event example.

Attention scores above the 85th percentile, computed across all layers and heads, are visualised to highlight the most significant interactions while reducing noise. By visualizing the attention maps, we can diagnose how attention guides the formation of feature representations, shedding light on the underlying mechanisms of the model. Fig. 5 shows the resulting image for the test set. Attention in the first layer displays higher magnitudes and broader distributions, capturing diverse relationships within the graph. In contrast, the second layer shows reduced magnitudes, indicating a refinement of features. This hierarchical approach reflects the model's process of first aggregating general dependencies and then fine-tuning these into task-relevant representations. The attention focus in head 2 is dominated by core features of $b1$, $b2$, whereas in head 1 the *energy*, *lepton* plays a more central role, with $b1$, $b2$ remaining of secondary relevance. This behaviour is consistently observed across both graph types. This is in excellent agreement with the physics expectations: the origin of the $b1, b2$ is a Higgs boson in signal that is not present in background, and as such is clearly distinctive; the missing transverse energy, in particular when related to the lepton object, is expected to be the next distinctive feature because it originates from the dark matter particles and the neutrino in signal, and only from the neutrino in background, with lepton and neutrino originating from the decay of the W boson (both in signal and background).

The above considerations are even better depicted when considering correctly classified signals (Fig. 6), where attention strongly concentrates on $b1$, $b2$, and *energy* nodes, highlighting their importance in distinguishing signal events. In contrast, for correctly classified backgrounds (Fig. 7), attention is more evenly distributed across nodes, reflecting the broader and less distinctive feature patterns typical of background events. Finally, we note how the additional jet node in 7-node graphs receives some attention in the second layer for signal events, indicating a refinement of features based on its correlation with the other jets in the event. More specifically, the attention score of $j3$ in 7-node graphs can be explained by the fact that the feature η is expected to have some distinctive power, being in average lower for background and larger for signal. The quantile of such extra jet is also an additional distinctive feature.

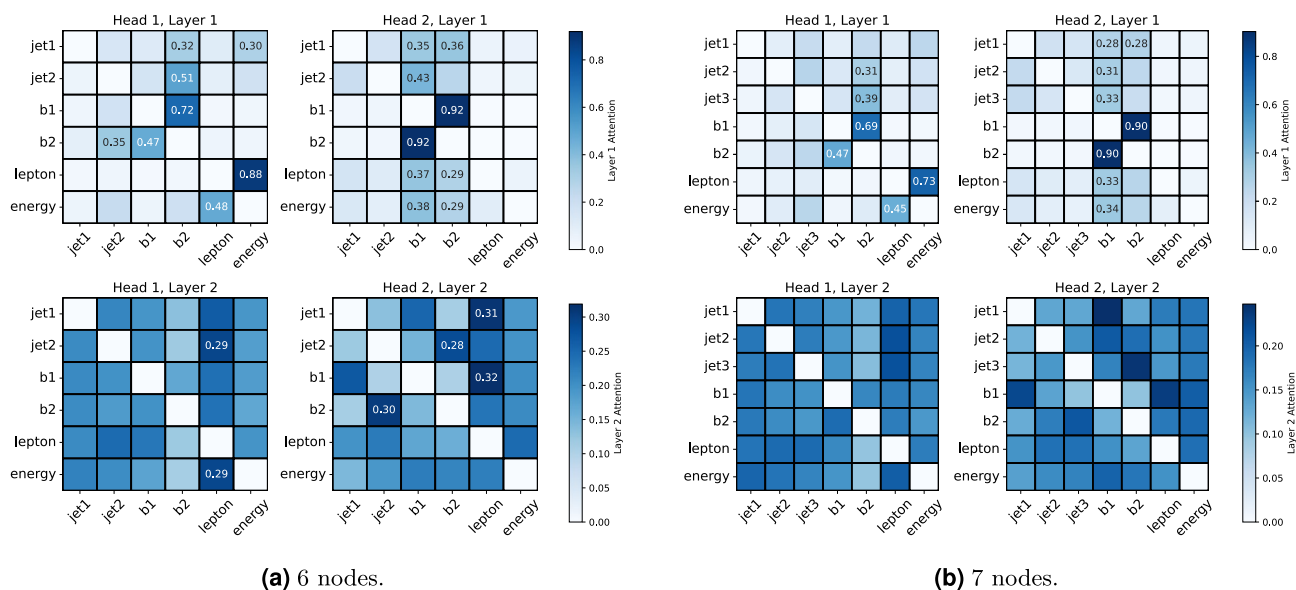


Fig. 5. Attention maps for test set. Each image shows the averaged attention scores for a specific attention head. The color scale ranges from white (low attention) to dark blue (high attention), visualizing only attention scores above the 85th percentile (computed across all layers and heads) to highlight the most significant interactions. The left column displays maps for 6-node graphs; the right column for 7-node graphs.

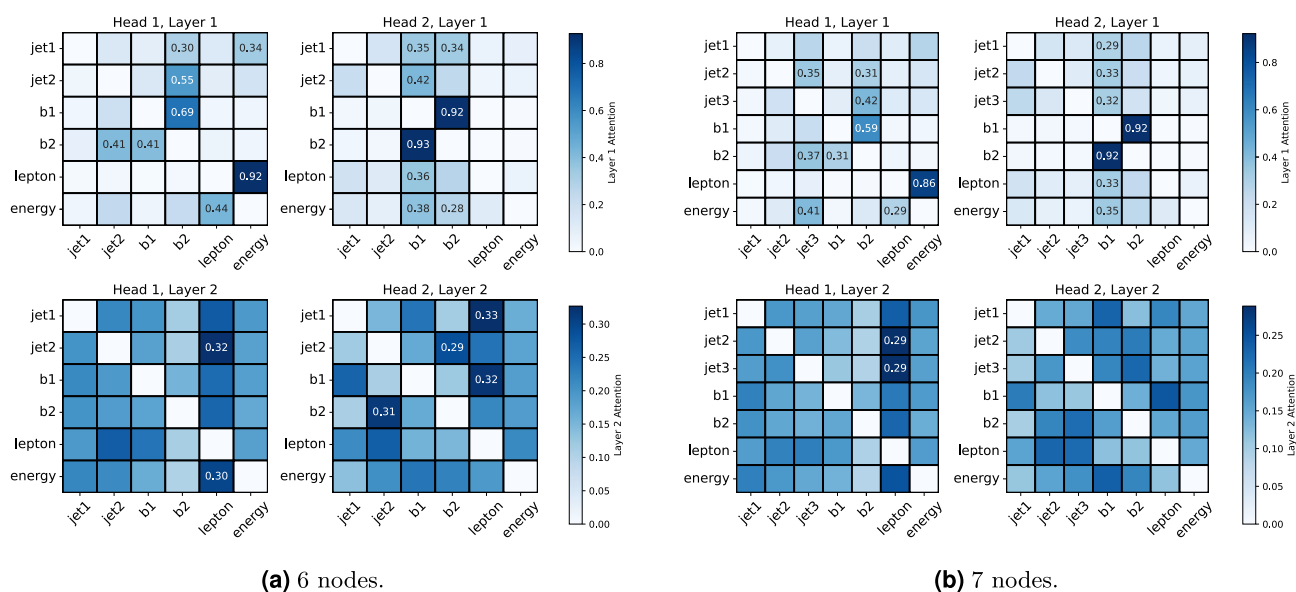


Fig. 6. Attention maps for correctly classified signals (true positives). Each image shows the averaged attention scores for a specific attention head. The color scale ranges from white (low attention) to dark blue (high attention), visualizing only attention scores above the 85th percentile (computed across all layers and heads) to highlight the most significant interactions. The left column displays maps for 6-node graphs; the right column for 7-node graphs.

Analysis of experts specialization

Unlike traditional neural network approaches that rely on a single monolithic network, MoE introduces a dynamic mechanism which can be helpful in processing and interpreting complex data. The key motivation behind employing MoE in our GT model is to understand how different expert networks specialize in processing distinct aspects of the interaction graph. We test node specialization which specifies how the model processes inputs by dividing the workload among its components. Each specialized expert becomes highly tuned to a certain type of input (in our case the type of node), allowing it to extract and represent the most relevant information from that subset. Using the same dataset of signal and background events, the specialization

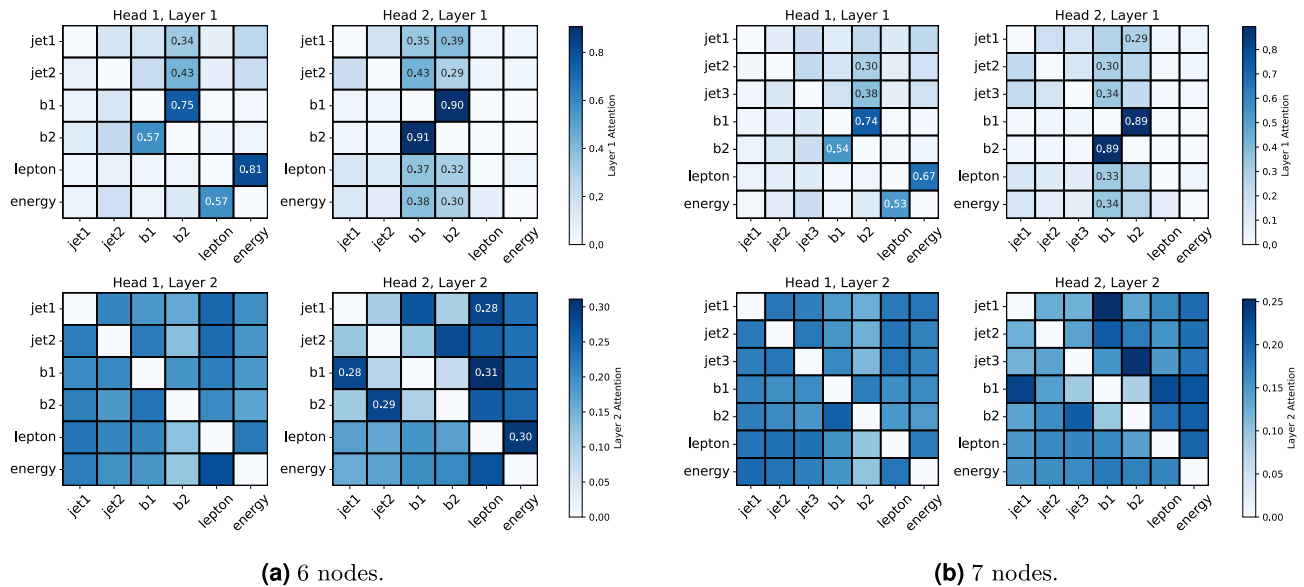


Fig. 7. Attention maps for correctly classified backgrounds (true negatives). Each image shows the averaged attention scores for a specific attention head. The color scale ranges from white (low attention) to dark blue (high attention), visualizing only attention scores above the 85th percentile (computed across all layers and heads) to highlight the most significant interactions. The left column displays maps for 6-node graphs; the right column for 7-node graphs.

patterns of the MoE block for layers 1 and 2 is studied. Fig. 8) highlights the distinct ways the model processes the features of different nodes. The MoE block of layer 1 appears to focus on extracting fundamental distinctions between node types, with strong specialization and minimal overlap between experts. The MoE block of layer 2, on the other hand, shows slightly more overlap among experts, suggesting that it operates at a higher level of abstraction, combining features and resolving correlations between particle types to refine the classification. In Tables 3 and 4, we present a summary of the specializations of the various experts, highlighting the nodes where the experts exhibit the highest activation. In more detail, for the MoE block of layer 1, Experts 2 and 4 focus on *lepton* and *energy* nodes, which are the quantities closely tied to the W boson. As highlighted before, in background events, the missing energy arises solely from the neutrino produced in the decay of the W boson whilst in signal events missing energy also originates from dark matter candidates. The specialization of Experts 2 and 4 indicates that the model identifies the combined behavior of *lepton* and missing energy features as a key discriminator. Together, these quantities encode differences between signal and background events, such as the overall energy imbalance or the transverse momentum carried away by undetected particles. The paired specialization of Experts 1 and 5 on the two b-jets suggests that the model learns to process b-jets as a correlated entity rather than treating them in isolation. This approach is consistent with the physics. Experts 3 and 6 demonstrate specialization in *jet1*, *jet2*, and to a lesser degree, *jet3*. This pattern can be explained by the structure of the dataset. For 6-node graphs, *jet1*, *jet2* are equivalent to *b1*, *b2*, hence the same specialization of Experts 1 and 5 is expected. For 7-node graphs, *jet3* acquires an added distinctive value. The fact that it is not consistently present across all events limits the opportunities for these experts to learn from its features. The MoE block of layer 2, on the other hand, shows slightly more overlap among experts. This may suggest that layer 2 operates at a higher level of abstraction—combining features and resolving correlations between particle types to refine the classification—the qualitative nature of this observation does not allow us to make a definitive quantitative claim.

Conclusion

In this work, we introduced a novel Mixture of Experts layer for graph data, aiming to enhance both explainability and performance in high-energy particle physics analysis. The proposed model, evaluated on a HEP dataset of simulated events from a search for dark matter in SUSY models published by the ATLAS experiment, achieved superior predictive accuracy compared to baseline models. This improvement underscores the effectiveness of leveraging expert subnetworks to address the task-specific complexities inherent in high-energy physics data. Beyond its strong predictive performance, the model made significant strides in addressing the critical need for explainability in high-energy physics. By visualizing attention maps and analyzing expert specializations, we identified key features and their influence on classification outcomes, ensuring that the model's decision-making aligns with established physical principles. This ability to trace predictions back to interpretable components not only enhances trust in the model but also provides a deeper understanding of the physics involved. Despite these achievements, the study has some limitations that warrant further exploration. The scalability of the MGT model to larger datasets or graphs with more nodes remains an open question, as does its ability to generalize across a wider range of high-energy physics tasks. Moreover, future validations of these results on other datasets and larger architectures will be critical in establishing their robustness and broader applicability. Future research

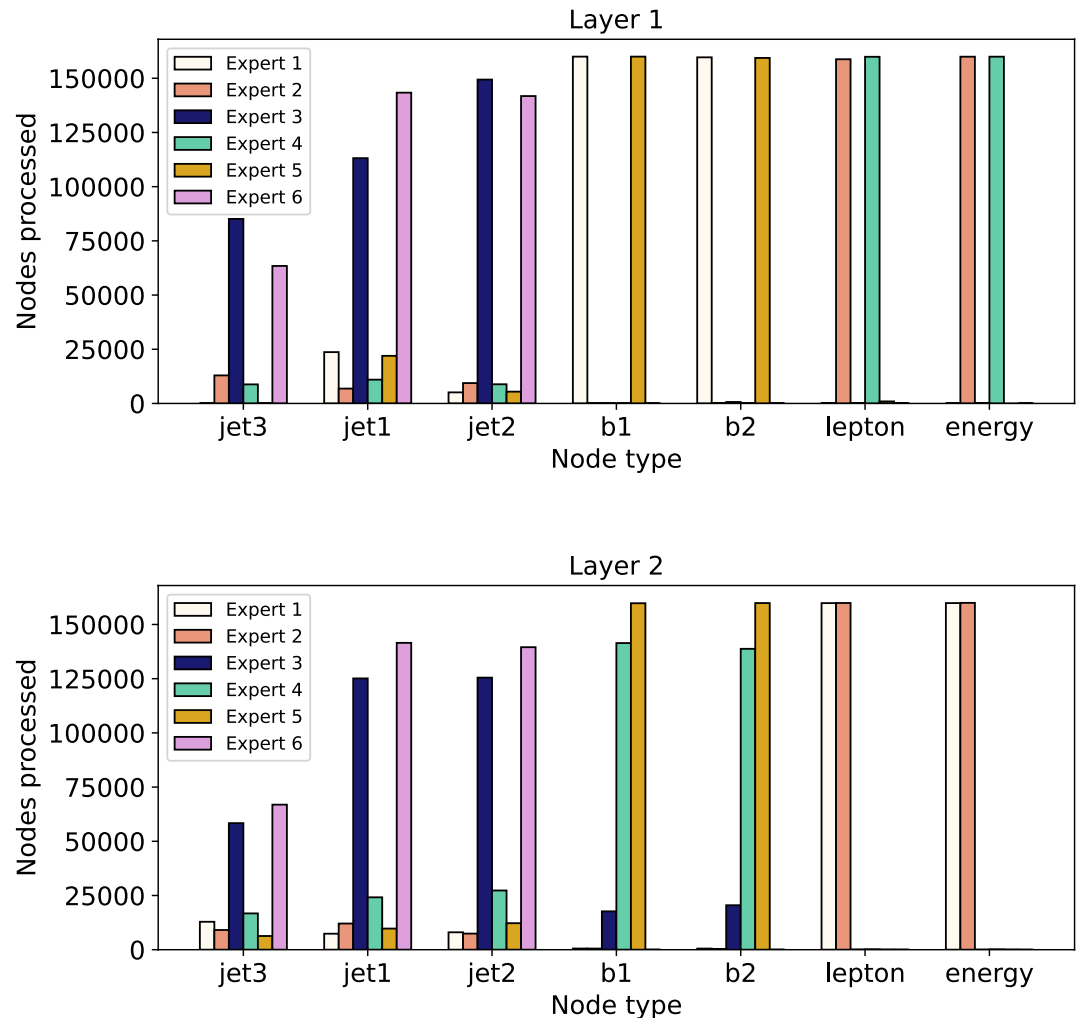


Fig. 8. Visualization of node specialization in the MoE architecture for layers 1 and 2. The bar charts show the number of nodes processed by each expert for different node types (*jet1, jet2, jet3, b1, b2, lepton, energy*).

Expert Number	Specialization	Physical Explanation
1, 5	<i>b1, b2</i>	Experts focus on the b-jets as a correlated entity. In signal events, these arise from Higgs boson decay, but in background events, they do not.
2, 4	<i>lepton, energy</i>	Experts specialize in detecting lepton and missing energy features, key indicators for distinguishing signal from background. In signal, missing energy also originates from dark matter particles, unlike the background.
3, 6	<i>jet1, jet2, jet3</i>	Equivalent to <i>b1</i> and <i>b2</i> in 6-node graphs, but in 7-node graphs, the additional jet (<i>jet3</i>) introduces distinct kinematic features that alter the specialization

Table 3. Expert Specialization for layer 1.

Expert Number	Specialization	Physical Explanation
1, 2	<i>lepton, energy</i>	Similar to Experts 2 and 4 in Layer 1
3, 6	<i>jet1, jet2, jet3</i>	Similar to Experts 3 and 6 in Layer 1, but with less specialization
4, 5	<i>b1, b2</i>	Similar to Experts 1 and 5 in Layer 1, but with less specialization

Table 4. Expert Specialization for layer 2.

could extend this framework to other types of datasets and particle physics phenomena, testing its adaptability and robustness across diverse scenarios. Furthermore, these results are particularly promising in relation to the application of advanced mechanistic explainability techniques, such as Sparse Autoencoders⁴⁴, to this type of data. The integration of emerging techniques for automated explainability, including the use of Large Language Models⁴⁵, could further enhance the accessibility and dynamism of the model's interpretations, amplifying its utility for scientific discovery. By combining high predictive performance with enhanced explainability, this work lays the groundwork for advancing explainable machine-learning methodologies in particle physics. Building on these foundations promises to unlock further potential in addressing the challenges of analyzing complex scientific data.

Data availability

The data used is public and released under Creative Commons Zero v1.0 Universal license. It is fully accessible via opendata <https://opendata.cern.ch/record/28100>, where a description of the content is included. The generation and simulation setup are fully detailed in the cited ATLAS paper (10.1007/JHEP12(2023)167). The code for the model is available at <https://github.com/DonatellaGenovese/Mixture-of-Expert-Graph-Transformer>.

Received: 23 January 2025; Accepted: 14 July 2025

Published online: 31 July 2025

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Acknowledgements

The work is partly funded by the European Union's CHIST-ERA programme under grant agreement CHIST-ERA-19-XAI-009 (MUCCA). DG is supported by PNRR MUR project PE0000013-FAIR. SS is partly funded by Sapienza grants RM1221816BD028D6 (DESMOS) and RG123188B3EF6A80 (CENTS). The authors would like to thank Dr Joseph Carmignani, former research assistant at the University of Liverpool supported by the MUCCA project, for his invaluable work on the SUSY ATLAS analysis and publication of the input datasets.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-12003-9>.

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